

2106.07890: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2106.07890_F00001	3	■ 0.985	$\mathcal{L}$	$\mathcal{L}$	defining the language syntax category $\mathcal{L}$ as a category enriched over $[0, 1]$. We then
2106.07890_F00002	4	■ 1.000	$\widehat{\mathcal{L}}$	$\widehat{\mathcal{L}}$	isfying a certain monotonicity condition. The category $\widehat{\mathcal{L}}$ of enriched copresheaves,
2106.07890_F00003	2	■ 0.814	$\{ \}^{\{+ \}}$	$+$	[VSP ⁺ 17, RNSS18, RWC ⁺ 18, B ⁺ 20]. Intuitively, the ability to continue a story
2106.07890_F00004	2	■ 0.814	$\mathrm{B}^{\{+ \}}$	B^+	[VSP ⁺ 17, RNSS18, RWC ⁺ 18, B ⁺ 20]. Intuitively, the ability to continue a story
2106.07890_F00005	3	■ 0.894	∞	∞	an isomorphism between $[0, 1]$ and the set of nonnegative extended reals $[0, \infty]$ to
2106.07890_F00006	5	■ 1.000	$x \rightarrow y$	$x \rightarrow y$	finite set of atomic symbols, where there is a morphism $x \rightarrow y$ whenever x is a
2106.07890_F00007	5	■ 1.000	x	x	finite set of atomic symbols, where there is a morphism $x \rightarrow y$ whenever x is a
2106.07890_F00008	5	■ 1.000	y	y	substring of y , that is, when the expression y is a <i>continuation</i> of x . If the finite set
2106.07890_F00009	5	■ 1.000	z	z	by transitivity: if x is a substring of y and y is a substring of z , then x is a substring
2106.07890_F00010	5	■ 0.997	$h^x := \operatorname{hom}(x, -)$	$h^x := \operatorname{hom}(x, -)$	expression x with the functor $h^x := \operatorname{hom}(x, -)$ whose value on an expression y is
2106.07890_F00011	5	■ 1.000	$*$	$*$	the one-point set $*$ if x is contained in y and is the empty set otherwise. The functor
2106.07890_F00012	5	■ 1.000	h^x	h^x	h^x is an example of a <i>copresheaf</i> , the name given to a functor from a given category
2106.07890_F00013	5	■ 0.946	$\widehat{\mathrm{C}} :=$	$\widehat{\mathrm{C}} :=$	shift in perspective is that the functor category of copresheaves $\widehat{\mathrm{C}} := \mathrm{Set}^{\mathrm{C}}$ on
2106.07890_F00014	5	■ 0.946	$^{\mathrm{C}}$	$^{\mathrm{C}}$	shift in perspective is that the functor category of copresheaves $\widehat{\mathrm{C}} := \mathrm{Set}^{\mathrm{C}}$ on
2106.07890_F00015	6	■ 0.709	$h^x \sqcup h^y$	$h^x \sqcup h^y$	$h^x \sqcup h^y$: $\mathrm{C} \rightarrow \mathrm{Set}$ is computed “pointwise.” So suppose C is the category of ex-
2106.07890_F00016	6	■ 0.925	$x =$	$x =$	pressions in language defined above, and let $x = \textit{red}$ and $y = \textit{blue}$. The coproduct
2106.07890_F00017	6	■ 0.925	$y =$	$y =$	pressions in language defined above, and let $x = \textit{red}$ and $y = \textit{blue}$. The coproduct
2106.07890_F00018	6	■ 1.000	$h^{\textit{red}} \sqcup h^{\textit{blue}}$	$h^{\textit{red}} \sqcup h^{\textit{blue}}$	$h^{\textit{red}} \sqcup h^{\textit{blue}}$ therefore maps an expression c to the set $h^{\textit{red}}(c) \sqcup h^{\textit{blue}}(c)$. This set is
2106.07890_F00019	6	■ 1.000	c	c	$h^{\textit{red}} \sqcup h^{\textit{blue}}$ therefore maps an expression c to the set $h^{\textit{red}}(c) \sqcup h^{\textit{blue}}(c)$. This set is
2106.07890_F00020	6	■ 1.000	$h^{\textit{red}}(c) \sqcup h^{\textit{blue}}(c)$	$h^{\textit{red}}(c) \sqcup h^{\textit{blue}}(c)$	$h^{\textit{red}} \sqcup h^{\textit{blue}}$ therefore maps an expression c to the set $h^{\textit{red}}(c) \sqcup h^{\textit{blue}}(c)$. This set is
2106.07890_F00021	6	■ 0.963	$h^x \times h^y : \mathrm{C} \rightarrow$	$h^x \times h^y : \mathrm{C} \rightarrow$	product, the product of representable copresheaves $h^x \times h^y : \mathrm{C} \rightarrow \mathrm{Set}$ is again a
2106.07890_F00022	6	■ 0.886	$h^{\textit{red}}(c) \times h^{\textit{blue}}(c)$	$h^{\textit{red}}(c) \times h^{\textit{blue}}(c)$	$h^{\textit{red}}(c) \times h^{\textit{blue}}(c)$, which is isomorphic to $*$ if c contains both <i>red</i> and <i>blue</i> and
2106.07890_F00023	6	■ 1.000	$h^{\textit{red}} \times h^{\textit{blue}}$	$h^{\textit{red}} \times h^{\textit{blue}}$	is the empty set otherwise. So the output of the functor $h^{\textit{red}} \times h^{\textit{blue}}$ is nonempty
2106.07890_F00024	6	■ 0.994	$\widehat{\mathrm{C}}$	$\widehat{\mathrm{C}}$	category of copresheaves $\widehat{\mathrm{C}}$ on a small category C . To say that $\widehat{\mathrm{C}}$ is <i>Cartesian closed</i>
2106.07890_F00025	6	■ 0.969	$[,] : \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$	$[,] : \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$	$[,] : \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$ called the <i>internal hom</i> that fits into an isomorphism $\widehat{\mathrm{C}}(F \times$
2106.07890_F00026	6	■ 0.969	$\widehat{\mathrm{C}}(F \times$	$\widehat{\mathrm{C}}(F \times$	$[,] : \widehat{\mathrm{C}} \times \widehat{\mathrm{C}} \rightarrow \widehat{\mathrm{C}}$ called the <i>internal hom</i> that fits into an isomorphism $\widehat{\mathrm{C}}(F \times$
2106.07890_F00027	6	■ 0.997	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$ for all copresheaves F, G and H . Here and throughout we
2106.07890_F00028	6	■ 0.997	F, G	F, G	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$ for all copresheaves F, G and H . Here and throughout we
2106.07890_F00029	6	■ 0.997	H	H	$G, H) \cong \widehat{\mathrm{C}}(F, [G, H])$ for all copresheaves F, G and H . Here and throughout we
2106.07890_F00030	6	■ 0.877	$\mathrm{A}(x, y)$	$\mathrm{A}(x, y)$	use the notation $\mathrm{A}(x, y)$ to denote the set of morphisms from objects x to y in a
2106.07890_F00031	6	■ 0.997	$[G, H]$	$[G, H]$	category A . The relevance of this new copresheaf $[G, H]$ can be seen in its kinship
2106.07890_F00032	6	■ 1.000	$x \leq y$	$x \leq y$	The usual notation in this setting is to write $x \leq y$ if there is a morphism from
2106.07890_F00033	6	■ 1.000	$\wedge$	$\wedge$	$x \rightarrow y$. The categorical product is denoted by $\wedge$ and the internal hom is denoted
2106.07890_F00034	6	■ 0.992	$x \wedge y \leq z$	$x \wedge y \leq z$	by $\Rightarrow$. So, in this notation, one has: $x \wedge y \leq z$ if and only if $x \leq (y \Rightarrow z)$ for
2106.07890_F00035	6	■ 0.992	$x \leq (y \Rightarrow z)$	$x \leq (y \Rightarrow z)$	by $\Rightarrow$. So, in this notation, one has: $x \wedge y \leq z$ if and only if $x \leq (y \Rightarrow z)$ for
2106.07890_F00036	6	■ 1.000	x, y, z	x, y, z	all elements x, y, z . Keep this analogy in mind as we dissect an example of the
2106.07890_F00037	7	■ 0.998	$h^c \times h^{\textit{red}}$	$h^c \times h^{\textit{red}}$	from the functor $h^c \times h^{\textit{red}}$ to the functor $h^{\textit{blue}}$. The data of a single natural trans-
2106.07890_F00038	7	■ 0.998	$h^{\textit{blue}}$	$h^{\textit{blue}}$	from the functor $h^c \times h^{\textit{red}}$ to the functor $h^{\textit{blue}}$. The data of a single natural trans-

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2106.07890_F00039	7	■ 0.991	$\left(h^{\{c\}} \times h^{\{\text{red}\}}\right)(d)$	$\left(h^c \times h^{\text{red}}\right)(d)$	that fit into a particular commutative square. Here, the domain $(h^c \times h^{\text{red}})(d)$ and
2106.07890_F00040	7	■ 0.991	$\left(h^{\{\text{blue}\}}\right)(d)$	$\left(h^{\text{blue}}\right)(d)$	the codomain $(h^{\text{blue}})(d)$ are either empty or a singleton, and so the data of a natural
2106.07890_F00041	7	■ 1.000	$\left[h^{\{\text{red}\}}, h^{\{\text{blue}\}}\right](c)$	$\left[h^{\text{red}}, h^{\text{blue}}\right](c)$	into the required commutative square. Therefore, $[h^{\text{red}}, h^{\text{blue}}](c)$ is either the empty
2106.07890_F00042	7	■ 0.983	$\emptyset$	$\emptyset$	set $\emptyset$ or the one-point set $*$.
2106.07890_F00043	7	■ 0.996	$h^{\{c\}} \times$	$h^c \times$	functions in (2). The product is computed pointwise, and thus the domain $(h^c \times$
2106.07890_F00044	7	■ 0.508	$h^{\{\text{red}\}}(d)=h^{\{c\}}(d) \times h^{\{\text{red}\}}(d)$	$h^{\text{red}}(d)=h^c(d) \times h^{\text{red}}(d)$	$h^{\text{red}})(d)=h^c(d) \times h^{\text{red}}(d)$ is isomorphic to the one-point set $*$ whenever d contains
2106.07890_F00045	7	■ 0.508	d	d	$h^{\text{red}})(d)=h^c(d) \times h^{\text{red}}(d)$ is isomorphic to the one-point set $*$ whenever d contains
2106.07890_F00046	7	■ 0.943	$h^{\{\text{blue}\}}(d)$	$h^{\text{blue}}(d)$	both c and red and is the empty set otherwise. The codomain $h^{\text{blue}}(d)$ is likewise
2106.07890_F00047	7	■ 0.991	$\left(h^{\{c\}} \times h^{\{\text{red}\}}\right)(d)=*$	$\left(h^c \times h^{\text{red}}\right)(d)=*$	text d that contains c and red but does not contain blue then $(h^c \times h^{\text{red}})(d)=*$ and
2106.07890_F00048	7	■ 1.000	$h^{\{\text{blue}\}}(d)=\emptyset$	$h^{\text{blue}}(d)=\emptyset$	$h^{\text{blue}}(d)=\emptyset$, hence there does not exist a function $(h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d)$ and
2106.07890_F00049	7	■ 1.000	$\left(h^{\{c\}} \times h^{\{\text{red}\}}\right)(d) \rightarrow h^{\{\text{blue}\}}(d)$	$\left(h^c \times h^{\text{red}}\right)(d) \rightarrow h^{\text{blue}}(d)$	$h^{\text{blue}}(d)=\emptyset$, hence there does not exist a function $(h^c \times h^{\text{red}})(d) \rightarrow h^{\text{blue}}(d)$ and
2106.07890_F00050	7	■ 0.936	$\left[h^{\{\text{red}\}}, h^{\{\text{blue}\}}\right]$	$\left[h^{\text{red}}, h^{\text{blue}}\right]$	and red as subtexts, also contains blue and so $[h^{\text{red}}, h^{\text{blue}}](\text{French flag})=*$. If
2106.07890_F00051	7	■ 0.936	$=*$	$=*$	and red as subtexts, also contains blue and so $[h^{\text{red}}, h^{\text{blue}}](\text{French flag})=*$. If
2106.07890_F00052	7	■ 0.996	$\left[h^{\{\text{red}\}}, h^{\{\text{blue}\}}\right]($	$\left[h^{\text{red}}, h^{\text{blue}}\right]($	$[h^{\text{red}}, h^{\text{blue}}](\text{ruby})=\emptyset$. One might say “ red implies blue in the context of French
2106.07890_F00053	7	■ 0.996	$)=\emptyset$	$)=\emptyset$	$[h^{\text{red}}, h^{\text{blue}}](\text{ruby})=\emptyset$. One might say “ red implies blue in the context of French
2106.07890_F00054	7	■ 1.000	$\left[h^{\{x\}}, h^{\{y\}}\right]: \mathrm{C} \rightarrow$	$[h^x, h^y]: \mathrm{C} \rightarrow$	In summary, the copresheaf $[h^x, h^y]: \mathrm{C} \rightarrow$ Set is given by
2106.07890_F00055	8	■ 0.781	$\mathrm{C}^{\mathrm{op}} \rightarrow$	$\mathrm{C}^{\mathrm{op}} \rightarrow$	a single morphism $x \rightarrow y$ if x is a subexpression of y . The functor $\mathrm{C}^{\mathrm{op}} \rightarrow \mathrm{Set}^{\mathrm{C}}$
2106.07890_F00056	8	■ 0.999	$h^{\{x\}}=\operatorname{hom}(x,-)$	$h^x=\operatorname{hom}(x,-)$	that maps an expression x to the representable functor $h^x=\operatorname{hom}(x,-)$ is called
2106.07890_F00057	8	■ 1.000	C^{op}	C^{op}	the <i>Yoneda embedding</i> and captures something of the meaning of x . Here C^{op} is
2106.07890_F00058	8	■ 0.994	$\mathrm{C}^{\mathrm{op}}(x, y):=\mathrm{C}(y, x)$	$\mathrm{C}^{\mathrm{op}}(x, y):=\mathrm{C}(y, x)$	morphisms are defined to be $\mathrm{C}^{\mathrm{op}}(x, y):=\mathrm{C}(y, x)$.) This is comparable to mak-
2106.07890_F00059	8	■ 0.994	$\mathrm{C}(x, y)$	$\mathrm{C}(x, y)$	simple. The set $\mathrm{C}(x, y)$ of morphisms from x to y is either the empty set or the
2106.07890_F00060	8	■ 0.793	$[0, 1]$	$[0, 1]$	2.1. Categories enriched over $[0, 1]$. A <i>preorder</i> is a set together with a reflex-
2106.07890_F00061	8	■ 1.000	a	a	elements of the set, with a single morphism from a to b if and only if a is related to
2106.07890_F00062	8	■ 1.000	b	b	elements of the set, with a single morphism from a to b if and only if a is related to
2106.07890_F00063	9	■ 0.957	$(\mathcal{V}, \leq, \otimes, 1)$	$(\mathcal{V}, \leq, \otimes, 1)$	Definition 1. A <i>commutative monoidal preorder</i> $(\mathcal{V}, \leq, \otimes, 1)$ is a preorder $(\mathcal{V}, \leq)$
2106.07890_F00064	9	■ 0.957	$(\mathcal{V}, \leq)$	$(\mathcal{V}, \leq)$	Definition 1. A <i>commutative monoidal preorder</i> $(\mathcal{V}, \leq, \otimes, 1)$ is a preorder $(\mathcal{V}, \leq)$
2106.07890_F00065	9	■ 0.994	$(\mathcal{V}, \otimes, 1)$	$(\mathcal{V}, \otimes, 1)$	and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2106.07890_F00066	9	■ 0.994	$x \otimes y \leq x^{\prime} \otimes y^{\prime}$	$x \otimes y \leq x' \otimes y'$	and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2106.07890_F00067	9	■ 0.994	$x \leq x^{\prime}$	$x \leq x'$	and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$
2106.07890_F00068	9	■ 1.000	$y \leq y^{\prime}$	$y \leq y'$	and $y \leq y'$.
2106.07890_F00069	9	■ 0.835	$(\mathcal{V}, \leq, \otimes, 1)$	$(\mathcal{V}, \leq, \otimes, 1)$	Definition 2. Let $(\mathcal{V}, \leq, \otimes, 1)$ be a commutative monoidal preorder. The data of
2106.07890_F00070	9	■ 1.000	$\mathcal{V}$	$\mathcal{V}$	a (small) $\mathcal{V}$ -enriched category, or simply a $\mathcal{V}$ -category, consists of a set of objects
2106.07890_F00071	9	■ 1.000	$\mathcal{C}$	$\mathcal{C}$	$\mathcal{C}$, and for every pair of objects x and y , there is an element $\mathcal{C}(x, y) \in \mathcal{V}$ called a
2106.07890_F00072	9	■ 1.000	$\mathcal{C}(x, y) \in \mathcal{V}$	$\mathcal{C}(x, y) \in \mathcal{V}$	$\mathcal{C}$, and for every pair of objects x and y , there is an element $\mathcal{C}(x, y) \in \mathcal{V}$ called a
2106.07890_F00073	9	■ 0.998	$x, y, z \in \mathcal{C}$	$x, y, z \in \mathcal{C}$	for all objects $x, y, z \in \mathcal{C}$.
2106.07890_F00074	9	■ 1.000	$[0, 1]:=\{x \in \mathbb{R}: 0 \leq x \leq 1\}$	$[0, 1]:=\{x \in \mathbb{R}: 0 \leq x \leq 1\}$	The unit interval $[0, 1]:=\{x \in \mathbb{R}: 0 \leq x \leq 1\}$ is a commutative monoidal
2106.07890_F00075	9	■ 0.923	$\leq$	$\leq$	with the usual $\leq$ relation being the preorder. The data of a $[0, 1]$ -category consists
2106.07890_F00076	9	■ 0.653	$(x, y) \mapsto \mathcal{C}(x, y)$	$(x, y) \mapsto \mathcal{C}(x, y)$	of a set of objects $\mathcal{C}$ and a $[0, 1]$ -valued function $(x, y) \mapsto \mathcal{C}(x, y)$ defined for every
2106.07890_F00077	9	■ 0.999	$x, y \in \mathcal{C}$	$x, y \in \mathcal{C}$	$x, y \in \mathcal{C}$ satisfying $\mathcal{C}(x, x)=1$ for every $x \in \mathcal{C}$ and $\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$ for
2106.07890_F00078	9	■ 0.999	$\mathcal{C}(x, x)=1$	$\mathcal{C}(x, x)=1$	$x, y \in \mathcal{C}$ satisfying $\mathcal{C}(x, x)=1$ for every $x \in \mathcal{C}$ and $\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$ for
2106.07890_F00079	9	■ 0.999	$x \in \mathcal{C}$	$x \in \mathcal{C}$	$x, y \in \mathcal{C}$ satisfying $\mathcal{C}(x, x)=1$ for every $x \in \mathcal{C}$ and $\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$ for
2106.07890_F00080	9	■ 0.999	$\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$	$\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$	$x, y \in \mathcal{C}$ satisfying $\mathcal{C}(x, x)=1$ for every $x \in \mathcal{C}$ and $\mathcal{C}(y, z) \mathcal{C}(x, y) \leq \mathcal{C}(x, z)$ for
2106.07890_F00081	9	■ 1.000	$[x, y] \in \mathcal{V}$	$[x, y] \in \mathcal{V}$	for every pair of elements x and y in $\mathcal{V}$ there is an element $[x, y] \in \mathcal{V}$, called the
2106.07890_F00082	9	■ 1.000	$\mathcal{V}(x, y)$	$\mathcal{V}(x, y)$	category enriched over itself by replacing the hom set $\mathcal{V}(x, y)$, which is either the

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2106.07890_F00083	10	■ 1.000	$[x, y]$	$[x, y]$	empty set or the one-point set, by the internal hom $[x, y]$, which is an object of $\mathcal{V}$.
2106.07890_F00084	10	■ 1.000	$(x, y) \mapsto [x, y]$	$(x, y) \mapsto [x, y]$	To see that the assignment $(x, y) \mapsto [x, y]$ does in fact make $\mathcal{V}$ a category enriched
2106.07890_F00085	10	■ 1.000	$1 \leq [x, x]$	$1 \leq [x, x]$	over itself, one needs to check that $1 \leq [x, x]$ and $[y, z] \otimes [x, y] \leq [x, z]$. The first
2106.07890_F00086	10	■ 1.000	$[y, z] \otimes [x, y] \leq [x, z]$	$[y, z] \otimes [x, y] \leq [x, z]$	over itself, one needs to check that $1 \leq [x, x]$ and $[y, z] \otimes [x, y] \leq [x, z]$. The first
2106.07890_F00087	10	■ 1.000	$1 \otimes x \leq x$	$1 \otimes x \leq x$	inequality follows from the fact that $1 \otimes x \leq x$ and Equation (5). One can check
2106.07890_F00088	10	■ 1.000	$[y, z] \leq [y, z]$	$[y, z] \leq [y, z]$	that $[y, z] \otimes [x, y] \leq [x, z]$ in two steps. First, $[y, z] \leq [y, z]$ and Equation (5) gives
2106.07890_F00089	10	■ 1.000	$[y, z] \otimes y \leq z$	$[y, z] \otimes y \leq z$	$[y, z] \otimes y \leq z$ and similarly, $[x, y] \otimes x \leq y$. Putting these two inequalities together
2106.07890_F00090	10	■ 1.000	$[x, y] \otimes x \leq y$	$[x, y] \otimes x \leq y$	$[y, z] \otimes y \leq z$ and similarly, $[x, y] \otimes x \leq y$. Putting these two inequalities together
2106.07890_F00091	10	■ 1.000	$[y, z] \otimes [x, y] \otimes x \leq z$	$[y, z] \otimes [x, y] \otimes x \leq z$	yields $[y, z] \otimes [x, y] \otimes x \leq z$ which implies $[y, z] \otimes [x, y] \leq [x, z]$. Much can
2106.07890_F00092	10	■ 1.000	$a \otimes b := a b$	$a \otimes b := ab$	<i>monoidal product</i> $a \otimes b := ab$ is the usual product of numbers and the internal
2106.07890_F00093	10	■ 1.000	$a, b \in [0, 1]$	$a, b \in [0, 1]$	<i>hom</i> is truncated division: for all $a, b \in [0, 1]$ define
2106.07890_F00094	10	■ 1.000	$a b \leq c$	$ab \leq c$	as an internal hom; that is, $ab \leq c$ if and only if $a \leq [b, c]$. There are two cases. If
2106.07890_F00095	10	■ 1.000	$a \leq [b, c]$	$a \leq [b, c]$	as an internal hom; that is, $ab \leq c$ if and only if $a \leq [b, c]$. There are two cases. If
2106.07890_F00096	10	■ 1.000	$c < b$	$c < b$	$c < b$ then $ab \leq c$ implies that $a \leq \frac{c}{b} = [b, c]$. If $b \leq c$ then $[b, c] = 1$ and $a \leq [b, c]$
2106.07890_F00097	10	■ 1.000	$a \leq \frac{c}{b} = [b, c]$	$a \leq \frac{c}{b} = [b, c]$	$c < b$ then $ab \leq c$ implies that $a \leq \frac{c}{b} = [b, c]$. If $b \leq c$ then $[b, c] = 1$ and $a \leq [b, c]$
2106.07890_F00098	10	■ 1.000	$b \leq c$	$b \leq c$	$c < b$ then $ab \leq c$ implies that $a \leq \frac{c}{b} = [b, c]$. If $b \leq c$ then $[b, c] = 1$ and $a \leq [b, c]$
2106.07890_F00099	10	■ 1.000	$[b, c] = 1$	$[b, c] = 1$	$c < b$ then $ab \leq c$ implies that $a \leq \frac{c}{b} = [b, c]$. If $b \leq c$ then $[b, c] = 1$ and $a \leq [b, c]$
2106.07890_F00100	10	—	$\square$	$\square$	$\square$
2106.07890_F00101	10	■ 1.000	$a, b, c \in [0, 1]$	$a, b, c \in [0, 1]$	all $a, b, c \in [0, 1]$ we have
2106.07890_F00102	10	■ 0.528	$\min\{b/a, 1\}$	$\min\{b/a, 1\}$	as the <i>closure</i> property of $[0, 1]$. We occasionally will write $\min\{b/a, 1\}$ when a
2106.07890_F00103	10	■ 1.000	$b/0 \geq 1$	$b/0 \geq 1$	might be zero. The reader should interpret $b/0 \geq 1$ for any $0 \leq b \leq 1$.
2106.07890_F00104	10	■ 1.000	$0 \leq b \leq 1$	$0 \leq b \leq 1$	might be zero. The reader should interpret $b/0 \geq 1$ for any $0 < b < 1$.
2106.07890_F00105	10	■ 1.000	$a b$	ab	Also, keep in mind that we use juxtaposition ab for ordinary multiplication of
2106.07890_F00106	10	■ 1.000	$a \otimes b$	$a \otimes b$	numbers. For our purposes, this is shorthand for the <i>monoidal</i> product $a \otimes b$. We
2106.07890_F00107	10	■ 1.000	$a \times b$	$a \times b$	two numbers $a, b \in [0, 1]$ the expression $a \times b$ means $\min\{a, b\}$. The <i>coproduct</i> is
2106.07890_F00108	10	■ 1.000	$\min\{a, b\}$	$\min\{a, b\}$	two numbers $a, b \in [0, 1]$ the expression $a \times b$ means $\min\{a, b\}$. The <i>coproduct</i> is
2106.07890_F00109	10	■ 0.999	$a \sqcup b$	$a \sqcup b$	given by the maximum and is denoted by $a \sqcup b$. So, when we are working in the
2106.07890_F00110	10	■ 0.722	$\mathrm{C}(x, y) = 1$	$C(x, y) = 1$	$[0, 1]$ by setting $C(x, y) = 1$ if there is a morphism between objects $x \rightarrow y$ and
2106.07890_F00111	10	■ 1.000	$\mathrm{C}(x, y) = 0$	$C(x, y) = 0$	$C(x, y) = 0$ otherwise. In fact, $2 := \{0, 1\}$, the set that contains zero and one
2106.07890_F00112	10	■ 1.000	$2 := \{0, 1\}$	$2 := \{0, 1\}$	$C(x, y) = 0$ otherwise. In fact, $2 := \{0, 1\}$, the set that contains zero and one
2106.07890_F00113	11	■ 1.000	$\pi(y \mid x)$	$\pi(y \mid x)$	where $\pi(y x)$ denotes the probability that expression y extends expression x . If x
2106.07890_F00114	11	■ 0.959	$\pi(y \mid x) = 0$	$\pi(y \mid x) = 0$	is not a subtext of y then necessarily $\pi(y x) = 0$.
2106.07890_F00115	11	■ 0.598	$\pi(x \mid x) =$	$\pi(x \mid x) =$	That $\mathcal{L}$ is indeed a category enriched over $[0, 1]$ follows from the fact that $\pi(x x) =$
2106.07890_F00116	11	■ 1.000	$\mathcal{D}$	$\mathcal{D}$	category $\mathcal{D}$ consisting of “density” operators. In this paper we construct a category
2106.07890_F00117	12	■ 0.963	$\mathcal{L} \rightarrow \widehat{\mathcal{L}}$	$\mathcal{L} \rightarrow \widehat{\mathcal{L}}$	defines an embedding $\mathcal{L} \rightarrow \widehat{\mathcal{L}}$. The reader may think of the $[0, 1]$ -enriched Yoneda
2106.07890_F00118	12	■ 1.000	$\mathcal{L} \rightarrow \mathcal{D}$	$\mathcal{L} \rightarrow \mathcal{D}$	embedding as factoring through the functor $\mathcal{L} \rightarrow \mathcal{D}$ defined in [BV20], as in the
2106.07890_F00119	12	■ 0.379	$\mathcal{D} \cdots \cdots \rightarrow \widehat{\mathcal{L}}$	$\mathcal{D} \cdots \cdots \rightarrow \widehat{\mathcal{L}}$	While technical details are needed to construct the third arrow $\mathcal{D} \cdots \cdots \rightarrow \widehat{\mathcal{L}}$, one
2106.07890_F00120	12	■ 0.940	$(\mathcal{V}, \otimes, \leq, 1)$	$(\mathcal{V}, \otimes, \leq, 1)$	preorder $(\mathcal{V}, \otimes, \leq, 1)$. An <i>enriched functor</i> $\mathcal{C} \rightarrow \mathcal{D}$ is a function $f: \mathcal{C} \rightarrow \mathcal{D}$ satis-
2106.07890_F00121	12	■ 0.940	$\mathcal{C} \rightarrow \mathcal{D}$	$\mathcal{C} \rightarrow \mathcal{D}$	preorder $(\mathcal{V}, \otimes, \leq, 1)$. An <i>enriched functor</i> $\mathcal{C} \rightarrow \mathcal{D}$ is a function $f: \mathcal{C} \rightarrow \mathcal{D}$ satis-
2106.07890_F00122	12	■ 0.940	$f: \mathcal{C} \rightarrow \mathcal{D}$	$f: \mathcal{C} \rightarrow \mathcal{D}$	preorder $(\mathcal{V}, \otimes, \leq, 1)$. An <i>enriched functor</i> $\mathcal{C} \rightarrow \mathcal{D}$ is a function $f: \mathcal{C} \rightarrow \mathcal{D}$ satis-
2106.07890_F00123	12	■ 1.000	$\mathcal{D} = \mathcal{V}$	$\mathcal{D} = \mathcal{V}$	In the case that $\mathcal{V}$ is closed, it is enriched over itself and we can take $\mathcal{D} = \mathcal{V}$ to
2106.07890_F00124	12	■ 1.000	$f: \mathcal{C} \rightarrow \mathcal{V}$	$f: \mathcal{C} \rightarrow \mathcal{V}$	preorder $\mathcal{V}$. An <i>enriched copresheaf</i> is a function $f: \mathcal{C} \rightarrow \mathcal{V}$ satisfying $\mathcal{C}(x, y) \leq$
2106.07890_F00125	12	■ 1.000	$\mathcal{C}(x, y) \leq$	$\mathcal{C}(x, y) \leq$	preorder $\mathcal{V}$. An <i>enriched copresheaf</i> is a function $f: \mathcal{C} \rightarrow \mathcal{V}$ satisfying $\mathcal{C}(x, y) \leq$
2106.07890_F00126	12	■ 1.000	$\mathcal{V}(fx, fy)$	$\mathcal{V}(fx, fy)$	$\mathcal{V}(fx, fy)$ for all objects x and y in $\mathcal{C}$.
2106.07890_F00127	12	■ 1.000	$\mathcal{D}^{\mathcal{C}}$	$\mathcal{D}^{\mathcal{C}}$	Now, we discuss how to make a $\mathcal{V}$ -enriched category $\mathcal{D}^{\mathcal{C}}$ whose objects are $\mathcal{V}$ -
2106.07890_F00128	12	■ 1.000	$\mathcal{D}^{\mathcal{C}}(f, g) \in \mathcal{V}$	$\mathcal{D}^{\mathcal{C}}(f, g) \in \mathcal{V}$	functors from $\mathcal{C}$ to $\mathcal{D}$. We need a hom object $\mathcal{D}^{\mathcal{C}}(f, g) \in \mathcal{V}$ between any two such
2106.07890_F00129	12	■ 1.000	$f, g: \mathcal{C} \rightarrow \mathcal{D}$	$f, g: \mathcal{C} \rightarrow \mathcal{D}$	functors $f, g: \mathcal{C} \rightarrow \mathcal{D}$. Formally, it is defined by an <i>end</i> , which is a particular kind

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2106.07890_F00130	13	■ 0.940	$\widehat{\mathcal{C}} := [0, 1]^{\mathcal{C}}$	$\widehat{\mathcal{C}} := [0, 1]^{\mathcal{C}}$	Lemma 2. If $\mathcal{C}$ is a category enriched over $[0, 1]$, then the category $\widehat{\mathcal{C}} := [0, 1]^{\mathcal{C}}$
2106.07890_F00131	13	■ 1.000	$f, g: \mathcal{C} \rightarrow [0, 1]$	$f, g: \mathcal{C} \rightarrow [0, 1]$	copresheaves $\boxed{f, g: \mathcal{C} \rightarrow [0, 1]}$ is given by the following infimum over all objects in
2106.07890_F00132	13	■ 1.000	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{1, gc/fc\}$	$\widehat{\mathcal{C}}(f, g) = \inf_{c \in \mathcal{C}} \{1, gc/fc\}$	is given by $\widehat{\mathcal{C}}(\boxed{f, g}) = \inf_{c \in \mathcal{C}} \boxed{1, gc/fc}$.
2106.07890_F00133	13	■ 1.000	$\mathcal{V} = [0, 1]$	$\mathcal{V} = [0, 1]$	<i>Proof.</i> Setting $\mathcal{V} = [0, 1]$ and also $\mathcal{D} = [0, 1]$, Equation (9) says what is required
2106.07890_F00134	13	■ 1.000	$\mathcal{D} = [0, 1]$	$\mathcal{D} = [0, 1]$	<i>Proof.</i> Setting $\mathcal{V} = [0, 1]$ and also $\mathcal{D} = [0, 1]$, Equation (9) says what is required
2106.07890_F00135	13	■ 1.000	$\mathcal{C} \rightarrow [0, 1]$	$\mathcal{C} \rightarrow [0, 1]$	for a function $\mathcal{C} \rightarrow [0, 1]$ to be an enriched functor. So, let $c, d \in \mathcal{C}$. We need to
2106.07890_F00136	13	■ 1.000	$c, d \in \mathcal{C}$	$c, d \in \mathcal{C}$	for a function $\mathcal{C} \rightarrow [0, 1]$ to be an enriched functor. So, let $c, d \in \mathcal{C}$. We need to
2106.07890_F00137	13	■ 1.000	$\mathcal{C}(c, d) \leq [\mathcal{C}(x, c), \mathcal{C}(x, d)]$	$\mathcal{C}(c, d) \leq [\mathcal{C}(x, c), \mathcal{C}(x, d)]$	That is, we need to check that $\boxed{\mathcal{C}(c, d) \leq [\mathcal{C}(x, c), \mathcal{C}(x, d)]}$, which by the closure
2106.07890_F00138	13	■ 1.000	$\mathcal{C}(c, d) \mathcal{C}(x, c) \leq \mathcal{C}(x, d)$	$\mathcal{C}(c, d) \mathcal{C}(x, c) \leq \mathcal{C}(x, d)$	condition in (7) is equivalent to $\mathcal{C}(c, d) \mathcal{C}(x, c) \leq \mathcal{C}(x, d)$, which is satisfied since
2106.07890_F00139	13	■ 0.990	$h^x :=$	$h^x :=$	Definition 7. For any object x in a $[0, 1]$ -category $\mathcal{C}$, we call the functor $h^x :=$
2106.07890_F00140	13	■ 0.999	$\mathcal{C}(x, -)$	$\mathcal{C}(x, -)$	$\mathcal{C}(x, -)$ the copresheaf represented by x . We say a copresheaf $f: \mathcal{C} \rightarrow [0, 1]$ is
2106.07890_F00141	13	■ 0.999	$f: \mathcal{C} \rightarrow [0, 1]$	$f: \mathcal{C} \rightarrow [0, 1]$	$\mathcal{C}(x, -)$ the copresheaf represented by x . We say a copresheaf $f: \mathcal{C} \rightarrow [0, 1]$ is
2106.07890_F00142	13	■ 1.000	$f = h^x$	$f = h^x$	representable if $\boxed{f = h^x}$ for some $x \in \mathcal{C}$.
2106.07890_F00143	13	■ 1.000	$x \mapsto h^x$	$x \mapsto h^x$	copresheaf. In this subsection, we see that the assignment $x \mapsto h^x$ defines an en-
2106.07890_F00144	13	■ 0.991	$\widehat{\mathcal{C}}(h^x, f) = f(x)$	$\widehat{\mathcal{C}}(h^x, f) = f(x)$	$\mathcal{C}$ and any $[0, 1]$ -copresheaf $f: \mathcal{C} \rightarrow [0, 1]$ we have $\boxed{\widehat{\mathcal{C}}(h^x, f) = f(x)}$.
2106.07890_F00145	13	■ 1.000	f	f	<i>Proof.</i> Fix an object $x \in \mathcal{C}$ and a copresheaf f . Since $\widehat{\mathcal{C}}(h^x, f) = \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\}$
2106.07890_F00146	13	■ 1.000	$\widehat{\mathcal{C}}(h^x, f) = \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\}$	$\widehat{\mathcal{C}}(h^x, f) = \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\}$	<i>Proof.</i> Fix an object $x \in \mathcal{C}$ and a copresheaf f . Since $\widehat{\mathcal{C}}(h^x, f) = \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\}$
2106.07890_F00147	13	■ 0.996	$c \in \mathcal{C}$	$c \in \mathcal{C}$	we have for any particular $\boxed{c \in \mathcal{C}}$ that $\widehat{\mathcal{C}}(h^x, f) \leq [h^x(c), fc]$. For $c = x$, we have
2106.07890_F00148	13	■ 0.996	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(c), fc]$	$\widehat{\mathcal{C}}(h^x, f) \leq [h^x(c), fc]$	we have for any particular $c \in \mathcal{C}$ that $\boxed{\widehat{\mathcal{C}}(h^x, f) \leq [h^x(c), fc]}$. For $c = x$, we have
2106.07890_F00149	13	■ 0.996	$c = x$	$c = x$	we have for any particular $c \in \mathcal{C}$ that $\widehat{\mathcal{C}}(h^x, f) \leq [h^x(c), fc]$. For $c = x$, we have
2106.07890_F00150	14	■ 0.738	$\mathcal{C}(x, c) \leq$	$\mathcal{C}(x, c) \leq$	On the other hand, since f is a $[0, 1]$ -functor from $\mathcal{C}$ to $[0, 1]$, we have $\mathcal{C}(x, c) \leq$
2106.07890_F00151	14	■ 0.971	$[fx, fc]$	$[fx, fc]$	$[fx, fc]$ for any $c \in \mathcal{C}$. By the closure of $[0, 1]$ the inequality $\mathcal{C}(x, c) \leq [fx, fc]$
2106.07890_F00152	14	■ 0.971	$\mathcal{C}(x, c) \leq [fx, fc]$	$\mathcal{C}(x, c) \leq [fx, fc]$	$[fx, fc]$ for any $c \in \mathcal{C}$. By the closure of $[0, 1]$ the inequality $\boxed{\mathcal{C}(x, c) \leq [fx, fc]}$
2106.07890_F00153	14	■ 1.000	$\mathcal{C}(x, c)fx \leq fc$	$\mathcal{C}(x, c)fx \leq fc$	is equivalent to $\mathcal{C}(x, c)fx \leq fc$ which in turn is equivalent to $fx \leq [\mathcal{C}(x, c), fc]$.
2106.07890_F00154	14	■ 1.000	$fx \leq [\mathcal{C}(x, c), fc]$	$fx \leq [\mathcal{C}(x, c), fc]$	is equivalent to $\mathcal{C}(x, c)fx \leq fc$ which in turn is equivalent to $\boxed{fx \leq [\mathcal{C}(x, c), fc]}$.
2106.07890_F00155	14	■ 1.000	$fx \leq \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\} =$	$fx \leq \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\} =$	Having $fx \leq [\mathcal{C}(x, c), fc]$ for every $c \in \mathcal{C}$ implies that $fx \leq \inf_{c \in \mathcal{C}} \{[h^x(c), fc]\} =$
2106.07890_F00156	14	■ 1.000	$\widehat{\mathcal{C}}(h^x, f)$	$\widehat{\mathcal{C}}(h^x, f)$	$\widehat{\mathcal{C}}(h^x, f)$ and the theorem is proved.
2106.07890_F00157	14	■ 0.900	$\mathcal{C}(y, x) = \widehat{\mathcal{C}}(h^x, h^y)$	$\mathcal{C}(y, x) = \widehat{\mathcal{C}}(h^x, h^y)$	Corollary 1. $\boxed{\mathcal{C}(y, x) = \widehat{\mathcal{C}}(h^x, h^y)}$ for all objects x, y in a $[0, 1]$ -category $\mathcal{C}$.
2106.07890_F00158	14	■ 0.900	x, y	x, y	Corollary 1. $\mathcal{C}(y, x) = \widehat{\mathcal{C}}(h^x, h^y)$ for all objects x, y in a $[0, 1]$ -category $\mathcal{C}$.
2106.07890_F00159	14	■ 1.000	$f = h^y$	$f = h^y$	<i>Proof.</i> Setting $f = h^y$ in Theorem 1 yields $\widehat{\mathcal{C}}(h^x, h^y) = h^y(x) = \mathcal{C}(y, x)$.
2106.07890_F00160	14	■ 1.000	$\widehat{\mathcal{C}}(h^x, h^y) = h^y(x) = \mathcal{C}(y, x)$	$\widehat{\mathcal{C}}(h^x, h^y) = h^y(x) = \mathcal{C}(y, x)$	<i>Proof.</i> Setting $f = h^y$ in Theorem 1 yields $\boxed{\widehat{\mathcal{C}}(h^x, h^y) = h^y(x) = \mathcal{C}(y, x)}$.
2106.07890_F00161	14	■ 1.000	$\mathcal{C}^{op} \rightarrow \widehat{\mathcal{C}}$	$\mathcal{C}^{op} \rightarrow \widehat{\mathcal{C}}$	functor $\mathcal{C}^{op} \rightarrow \widehat{\mathcal{C}}$, embedding $\mathcal{C}^{op}$ as an enriched subcategory of $\widehat{\mathcal{C}}$.
2106.07890_F00162	14	■ 1.000	$\mathcal{C}^{op}$	$\mathcal{C}^{op}$	functor $\mathcal{C}^{op} \rightarrow \widehat{\mathcal{C}}$, embedding $\mathcal{C}^{op}$ as an enriched subcategory of $\widehat{\mathcal{C}}$.
2106.07890_F00163	14	■ 1.000	$\widehat{\mathcal{C}}$	$\widehat{\mathcal{C}}$	functor $\mathcal{C}^{op} \rightarrow \widehat{\mathcal{C}}$, embedding $\mathcal{C}^{op}$ as an enriched subcategory of $\widehat{\mathcal{C}}$.
2106.07890_F00164	14	■ 1.000	op	op	The op in $\mathcal{C}^{op}$ stands for “opposite” and is there because the assignment $x \mapsto h^x$
2106.07890_F00165	14	■ 1.000	$\mathcal{C}^{op}$	$\mathcal{C}^{op}$	The op in $\mathcal{C}^{op}$ stands for “opposite” and is there because the assignment $x \mapsto h^x$
2106.07890_F00166	14	■ 1.000	Y	Y	3.2. The semantic category $\widehat{\mathcal{L}}$. Often, when a mathematical object Y has nice
2106.07890_F00167	14	■ 0.999	$\{X \rightarrow Y\}$	$\{X \rightarrow Y\}$	structure, the set of functions $\{X \rightarrow Y\}$ from a fixed X has nice struture also,

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2106.07890_F00168	14	■ 0.999	X	X	structure, the set of functions $\{X \rightarrow Y\}$ from a fixed X has nice struture also,
2106.07890_F00169	14	■ 1.000	$\widehat{\mathrm{C}}=[0,1]^{\mathrm{C}}$	$\widehat{C} = [0, 1]^C$	enriched category $\mathcal{C}$, the category $\widehat{C} = [0, 1]^C$ of copresheaves on $\mathcal{C}$, as a category
2106.07890_F00170	14	■ 0.674	$\widehat{\mathrm{L}}:=[0,1]^{\mathrm{L}}$	$\widehat{\mathcal{L}} := [0, 1]^{\mathcal{L}}$	$\widehat{\mathcal{L}} := [0, 1]^{\mathcal{L}}$ is the $[0, 1]$ -category of $[0, 1]$ -enriched copresheaves on the $[0, 1]$ -
2106.07890_F00171	14	■ 1.000	$h^x:=\mathrm{L}(x,-)$	$h^x := \mathcal{L}(x, -)$	For each object x in $\mathcal{L}$, the representable copresheaf $h^x := \mathcal{L}(x, -)$ is given by
2106.07890_F00172	14	■ 1.000	$x \leq c$	$x \leq c$	where we use “ $x \leq c$ ” as shorthand for “ x is contained as a subtext of c .”
2106.07890_F00173	15	■ 1.000	$\mathrm{L}(x,y)=a \neq 0$	$\mathcal{L}(x, y) = a \neq 0$	tation here also. Suppose the text y extends the text x and $\mathcal{L}(x, y) = a \neq 0$. We
2106.07890_F00174	15	■ 1.000	h^y	h^y	resents the varying potential contexts in which x might appear, and h^y represents
2106.07890_F00175	15	■ 0.850	C	C	the way (co)limits in the functor category Set^C when C is an ordinary category are
2106.07890_F00176	16	■ 0.666	$F: \mathrm{J} \rightarrow \mathrm{C}$	$F: J \rightarrow C$	category and let C be any category. Recall that the <i>limit</i> of a diagram $F: J \rightarrow C$,
2106.07890_F00177	16	■ 0.965	$\lim F$	$\lim F$	if it exists, is an object $\lim F$ in C together with the following isomorphism of
2106.07890_F00178	16	■ 0.999	Z	Z	So for each object Z in C there is a bijection $C(Z, \lim F) \cong \mathrm{Set}^J(*, C(Z, F))$
2106.07890_F00179	16	■ 0.999	$\mathrm{C}(Z, \lim F) \cong \operatorname{Set}^{\mathrm{J}}(*, \mathrm{C}(Z, F))$	$C(Z, \lim F) \cong \mathrm{Set}^J(*, C(Z, F))$	So for each object Z in C there is a bijection $C(Z, \lim F) \cong \mathrm{Set}^J(*, C(Z, F))$
2106.07890_F00180	16	■ 0.647	(Z, F)	(Z, F)	for each object Z in C , a natural transformation from $*$ to $C(Z, F)$ consists of a
2106.07890_F00181	16	■ 1.000	$*=*(i) \rightarrow \mathrm{C}(Z, F i)$	$* = *(i) \rightarrow C(Z, Fi)$	function $* = *(i) \rightarrow C(Z, Fi)$ for each object i in J . This simply picks out a
2106.07890_F00182	16	■ 1.000	i	i	function $* = *(i) \rightarrow C(Z, Fi)$ for each object i in J . This simply picks out a
2106.07890_F00183	16	■ 0.755	$Z \rightarrow F i$	$Z \rightarrow Fi$	morphism $Z \rightarrow Fi$ in C , and these morphisms comprise the legs of the limit cone
2106.07890_F00184	16	■ 0.999	$W: \mathrm{J} \rightarrow \mathrm{V}$	$W: \mathcal{J} \rightarrow \mathcal{V}$	placed with a so-called $\mathcal{V}$ -functor of weights $W: \mathcal{J} \rightarrow \mathcal{V}$, where $\mathcal{V}$ is the base cat-
2106.07890_F00185	16	■ 1.000	J	$\mathcal{J}$	egory over which enrichment takes place and where $\mathcal{J}$ is an indexing $\mathcal{V}$ -category.
2106.07890_F00186	16	■ 1.000	E	$\mathcal{E}$	Definition 9. Let $\mathcal{V}$ be a closed commutative monoidal category and let $\mathcal{J}$ and $\mathcal{E}$
2106.07890_F00187	16	■ 1.000	$F: \mathrm{J} \rightarrow \mathrm{E}$	$F: \mathcal{J} \rightarrow \mathcal{E}$	be $\mathcal{V}$ -categories. Given a $\mathcal{V}$ -functor $F: \mathcal{J} \rightarrow \mathcal{E}$ and a $\mathcal{V}$ -functor $W: \mathcal{J} \rightarrow \mathcal{V}$,
2106.07890_F00188	16	■ 1.000	F	F	the <i>weighted limit</i> of F , if it exists, is an object $\lim_W F$ of $\mathcal{E}$ together with the
2106.07890_F00189	16	■ 1.000	$\lim_{\mathrm{W}} F$	$\lim_W F$	the <i>weighted limit</i> of F , if it exists, is an object $\lim_W F$ of $\mathcal{E}$ together with the
2106.07890_F00190	16	■ 1.000	$\mathrm{E} \rightarrow \mathrm{V}$	$\mathcal{E} \rightarrow \mathcal{V}$	following isomorphism of $\mathcal{V}$ -functors $\mathcal{E} \rightarrow \mathcal{V}$,
2106.07890_F00191	17	■ 1.000	$\mathrm{E} \left(Z, \lim_{\mathrm{W}} F \right) \cong$	$\mathcal{E} \left(Z, \lim_W F \right) \cong$	So in other words, for each object Z in $\mathcal{E}$ there is an isomorphism $\mathcal{E}(Z, \lim_W F) \cong$
2106.07890_F00192	17	■ 1.000	$\mathrm{V}^{\mathrm{J}}(W, \mathrm{E}(Z, F))$	$\mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(Z, F))$	$\mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(Z, F))$ of objects in $\mathcal{V}$. The idea behind a weighted colimit is analogous.
2106.07890_F00193	17	■ 1.000	$W: \mathrm{J}^{\mathrm{op}} \rightarrow \mathrm{V}$	$W: \mathcal{J}^{\mathrm{op}} \rightarrow \mathcal{V}$	Given a $\mathcal{V}$ -functor $F: \mathcal{J} \rightarrow \mathcal{E}$ and a $\mathcal{V}$ -functor of weights $W: \mathcal{J}^{\mathrm{op}} \rightarrow \mathcal{V}$, the
2106.07890_F00194	17	■ 0.997	$\operatorname{colim}_{\mathrm{W}} F$	$\operatorname{colim}_W F$	<i>weighted colimit</i> of F is an object $\operatorname{colim}_W F$ of $\mathcal{E}$ together with an isomorphism
2106.07890_F00195	17	■ 1.000	$\mathrm{E}=\widehat{\mathrm{L}}:=[0,1]^{\mathrm{L}}$	$\mathcal{E} = \widehat{\mathcal{L}} := [0, 1]^{\mathcal{L}}$	case when $\mathcal{V} = [0, 1]$, when $\mathcal{E} = \widehat{\mathcal{L}} := [0, 1]^{\mathcal{L}}$, and when the indexing category $\mathcal{J}$
2106.07890_F00196	17	■ 1.000	$\mathrm{J}(i, j)=\delta_{i j}$	$\mathcal{J}(i, j) = \delta_{ij}$	setting $\mathcal{J}(i, j) = \delta_{ij}$ for $i, j \in \{1, 2\}$. To begin, fix a functor of weights $W: \mathcal{J} \rightarrow$
2106.07890_F00197	17	■ 1.000	$i, j \in\{1,2\}$	$i, j \in \{1, 2\}$	setting $\mathcal{J}(i, j) = \delta_{ij}$ for $i, j \in \{1, 2\}$. To begin, fix a functor of weights $W: \mathcal{J} \rightarrow$
2106.07890_F00198	17	■ 1.000	$W: \mathrm{J} \rightarrow$	$W: \mathcal{J} \rightarrow$	setting $\mathcal{J}(i, j) = \delta_{ij}$ for $i, j \in \{1, 2\}$. To begin, fix a functor of weights $W: \mathcal{J} \rightarrow$
2106.07890_F00199	17	■ 0.516	$w_1:=W(1)$	$w_1 := W(1)$	$[0, 1]$. This is nothing more than a choice of two numbers $w_1 := W(1)$ and $w_2 :=$
2106.07890_F00200	17	■ 0.516	$w_2:=$	$w_2 :=$	$[0, 1]$. This is nothing more than a choice of two numbers $w_1 := W(1)$ and $w_2 :=$
2106.07890_F00201	17	■ 1.000	$W(2)$	$W(2)$	$W(2)$. Further, for a fixed pair of copresheaves $f, g: \mathcal{L} \rightarrow [0, 1]$, define $F: \mathcal{J} \rightarrow$
2106.07890_F00202	17	■ 1.000	$f, g: \mathrm{L} \rightarrow[0,1]$	$f, g: \mathcal{L} \rightarrow [0, 1]$	$W(2)$. Further, for a fixed pair of copresheaves $f, g: \mathcal{L} \rightarrow [0, 1]$, define $F: \mathcal{J} \rightarrow$
2106.07890_F00203	17	■ 1.000	$F: \mathrm{J} \rightarrow$	$F: \mathcal{J} \rightarrow$	$W(2)$. Further, for a fixed pair of copresheaves $f, g: \mathcal{L} \rightarrow [0, 1]$, define $F: \mathcal{J} \rightarrow$
2106.07890_F00204	17	■ 0.801	$f:=F(1)$	$f := F(1)$	$\widehat{\mathcal{L}}$ to be the $[0, 1]$ -functor with $f := F(1)$ and $g := F(2)$.
2106.07890_F00205	17	■ 0.801	$g:=F(2)$	$g := F(2)$	$\widehat{\mathcal{L}}$ to be the $[0, 1]$ -functor with $f := F(1)$ and $g := F(2)$.
2106.07890_F00206	17	■ 1.000	W	W	Definition 10. Denote the weighted limit of F with respect to the weight W by
2106.07890_F00207	17	■ 0.996	$\left(w_1, f\right) \times\left(w_2, g\right): \mathrm{L} \rightarrow[0,1]$	$(w_1, f) \times (w_2, g): \mathcal{L} \rightarrow [0, 1]$	Theorem 2. The weighted limit $(w_1, f) \times (w_2, g): \mathcal{L} \rightarrow [0, 1]$ is given by $c \mapsto$
2106.07890_F00208	17	■ 0.996	$c \mapsto$	$c \mapsto$	Theorem 2. The weighted limit $(w_1, f) \times (w_2, g): \mathcal{L} \rightarrow [0, 1]$ is given by $c \mapsto$
2106.07890_F00209	17	■ 0.999	$\min \left\{\frac{f c}{w_1}, \frac{g c}{w_2}, 1\right\}$	$\min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$	$\min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$.

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2106.07890_F00210	17	■ 1.000	$c \mapsto \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$	$c \mapsto \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$	<i>Proof.</i> To check that $c \mapsto \min \left\{ \frac{fc}{w_1}, \frac{gc}{w_2}, 1 \right\}$ satisfies the universal property of the
2106.07890_F00211	17	■ 1.000	$Z: \mathcal{L} \rightarrow [0, 1]$	$Z: \mathcal{L} \rightarrow [0, 1]$	weighted limit, let $Z: \mathcal{L} \rightarrow [0, 1]$ be any copresheaf and look at Equation (15)
2106.07890_F00212	17	■ 1.000	$\widehat{\mathcal{L}}(Z, F)$	$\widehat{\mathcal{L}}(Z, F)$	pression $\widehat{\mathcal{L}}(Z, F)$, which is a functor $J \rightarrow [0, 1]$. Evaluating at i in J yields the
2106.07890_F00213	17	■ 1.000	$J \rightarrow [0, 1]$	$J \rightarrow [0, 1]$	pression $\widehat{\mathcal{L}}(Z, F)$, which is a functor $J \rightarrow [0, 1]$. Evaluating at i in J yields the
2106.07890_F00214	17	■ 1.000	J	J	pression $\widehat{\mathcal{L}}(Z, F)$, which is a functor $J \rightarrow [0, 1]$. Evaluating at i in J yields the
2106.07890_F00215	17	■ 1.000	$[0, 1]^{\mathcal{J}}$	$[0, 1]^{\mathcal{J}}$	Now, the right hand side of Equation (17), as a hom-object in $[0, 1]^{\mathcal{J}}$, is the mini-
2106.07890_F00216	17	■ 0.894	$f=F(1)$	$f = F(1)$	mum over the objects of $\mathcal{J}$ which are 1 and 2. Using $f = F(1)$ and $g = F(2)$, we
2106.07890_F00217	17	■ 0.894	$g=F(2)$	$g = F(2)$	mum over the objects of $\mathcal{J}$ which are 1 and 2. Using $f = F(1)$ and $g = F(2)$, we
2106.07890_F00218	18	■ 1.000	$\mathcal{L}(c, d) \leq \left[\lim_W F(c), \lim_W F(d) \right]$	$\mathcal{L}(c, d) \leq \left[\lim_W F(c), \lim_W F(d) \right]$	copresheaf; that is, that $\mathcal{L}(c, d) \leq [\lim_W F(c), \lim_W F(d)]$, or equivalently $\mathcal{L}(c, d) \lim_W F(c) \leq$
2106.07890_F00219	18	■ 1.000	$\mathcal{L}(c, d) \lim_W F(c) \leq$	$\mathcal{L}(c, d) \lim_W F(c) \leq$	copresheaf; that is, that $\mathcal{L}(c, d) \leq [\lim_W F(c), \lim_W F(d)]$, or equivalently $\mathcal{L}(c, d) \lim_W F(c) \leq$
2106.07890_F00220	18	■ 1.000	$\lim_W F(d)$	$\lim_W F(d)$	$\lim_W F(d)$, for all $c, d \in \mathcal{L}$. This desired inequality follows from the simple ob-
2106.07890_F00221	18	■ 1.000	$c, d \in \mathcal{L}$	$c, d \in \mathcal{L}$	$\lim_W F(d)$, for all $c, d \in \mathcal{L}$. This desired inequality follows from the simple ob-
2106.07890_F00222	18	■ 0.967	g	g	where the first inequality follows from the assumption that f and g are $[0, 1]$ -
2106.07890_F00223	18	■ 1.000	$(w_1, f) \times (w_2, g)$	$(w_1, f) \times (w_2, g)$	let's look closer at the weighted product $(w_1, f) \times (w_2, g)$ in the case f and g are
2106.07890_F00224	18	■ 1.000	$w_1, w_2 \in [0, 1]$	$w_1, w_2 \in [0, 1]$	$w_1, w_2 \in [0, 1]$. The weighted product $(w_1, h^x) \times (w_2, h^y): \mathcal{L} \rightarrow [0, 1]$ assigns to
2106.07890_F00225	18	■ 1.000	$(w_1, h^x) \times (w_2, h^y): \mathcal{L} \rightarrow [0, 1]$	$(w_1, h^x) \times (w_2, h^y): \mathcal{L} \rightarrow [0, 1]$	$w_1, w_2 \in [0, 1]$. The weighted product $(w_1, h^x) \times (w_2, h^y): \mathcal{L} \rightarrow [0, 1]$ assigns to
2106.07890_F00226	18	■ 1.000	$\pi(c \mid x) \leq w_1$	$\pi(c \mid x) \leq w_1$	expressions x and y are jointly contained within c , and whether $\pi(c \mid x) \leq w_1$ and
2106.07890_F00227	18	■ 0.996	$\pi(c \mid y) \leq w_2$	$\pi(c \mid y) \leq w_2$	$\pi(c \mid y) \leq w_2$.
2106.07890_F00228	18	■ 1.000	$w_1 h^x(c) \times w_2 h^y(c)$	$w_1 h^x(c) \times w_2 h^y(c)$	(19) increase and thus contribute less to the value of $w_1 h^x(c) \times w_2 h^y(c)$, which is a
2106.07890_F00229	18	■ 1.000	$(w_1, h^x) \times (w_2, h^y)$	$(w_1, h^x) \times (w_2, h^y)$	minimum. The copresheaf $(w_1, h^x) \times (w_2, h^y)$ captures something like a weighted
2106.07890_F00230	18	■ 1.000	$w_1 = w_2 = 1$	$w_1 = w_2 = 1$	It's worth looking closer when the weights are $w_1 = w_2 = 1$. In this case,
2106.07890_F00231	18	■ 1.000	$f \times g$	$f \times g$	we denote the weighted product simply by $f \times g$. This weighted product when
2106.07890_F00232	18	■ 1.000	$f, g, h: \mathcal{L} \rightarrow [0, 1]$	$f, g, h: \mathcal{L} \rightarrow [0, 1]$	Lemma 4. For any copresheaves $f, g, h: \mathcal{L} \rightarrow [0, 1]$ we have
2106.07890_F00233	19	■ 0.991	$W: \mathcal{J}^{\text{op}} \rightarrow [0, 1]$	$W: \mathcal{J}^{\text{op}} \rightarrow [0, 1]$	with two objects. Fix a functor of weights $W: \mathcal{J}^{\text{op}} \rightarrow [0, 1]$ setting $w_1 := W(1)$
2106.07890_F00234	19	■ 1.000	$w_2 := W(2)$	$w_2 := W(2)$	and $w_2 := W(2)$ and a diagram $F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$ with $f := F(1)$ and $g := F(2)$.
2106.07890_F00235	19	■ 1.000	$F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$	$F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$	and $w_2 := W(2)$ and a diagram $F: \mathcal{J} \rightarrow \widehat{\mathcal{L}}$ with $f := F(1)$ and $g := F(2)$.
2106.07890_F00236	19	■ 0.998	$(w_1, f) \sqcup (w_2, g): \mathcal{L} \rightarrow [0, 1]$	$(w_1, f) \sqcup (w_2, g): \mathcal{L} \rightarrow [0, 1]$	Theorem 3. The weighted colimit $(w_1, f) \sqcup (w_2, g): \mathcal{L} \rightarrow [0, 1]$ is given by $c \mapsto$
2106.07890_F00237	19	■ 1.000	$\max \{w_1 f c, w_2 g c\}$	$\max \{w_1 f c, w_2 g c\}$	$\max \{w_1 f c, w_2 g c\}$.
2106.07890_F00238	19	■ 0.539	$\mathcal{J} = \mathcal{J}^{\text{op}}$	$\mathcal{J} = \mathcal{J}^{\text{op}}$	<i>Proof.</i> Note that $\mathcal{J} = \mathcal{J}^{\text{op}}$ since there are no nonidentity morphisms in $\mathcal{J}$. Let
2106.07890_F00239	19	■ 1.000	$\widehat{\mathcal{L}}(F-, Z)$	$\widehat{\mathcal{L}}(F-, Z)$	Evaluating $\widehat{\mathcal{L}}(F-, Z)$ at $i = 1, 2$ yields
2106.07890_F00240	19	■ 1.000	$i = 1, 2$	$i = 1, 2$	Evaluating $\widehat{\mathcal{L}}(F-, Z)$ at $i = 1, 2$ yields
2106.07890_F00241	20	■ 0.912	$c \mapsto \max \{w_1 f c, w_2 g c\}$	$c \mapsto \max \{w_1 f c, w_2 g c\}$	It remains to check that the assignment $c \mapsto \max \{w_1 f c, w_2 g c\}$ is indeed a $[0, 1]$ -
2106.07890_F00242	20	■ 1.000	$\mathcal{L}(c, d) \leq [\text{colim}_W F(c), \text{colim}_W F(d)]$	$\mathcal{L}(c, d) \leq [\text{colim}_W F(c), \text{colim}_W F(d)]$	copresheaf; that is, that $\mathcal{L}(c, d) \leq [\text{colim}_W F(c), \text{colim}_W F(d)]$, or equivalently
2106.07890_F00243	20	■ 1.000	$\mathcal{L}(c, d) \text{colim}_W F(c) \leq \text{colim}_W F(d)$	$\mathcal{L}(c, d) \text{colim}_W F(c) \leq \text{colim}_W F(d)$	$\mathcal{L}(c, d) \text{colim}_W F(c) \leq \text{colim}_W F(d)$, for all $c, d \in \mathcal{L}$. This desired inequality
2106.07890_F00244	20	■ 1.000	$(w_1, f) \sqcup (w_2, g)$	$(w_1, f) \sqcup (w_2, g)$	The support of the copresheaf $(w_1, f) \sqcup (w_2, g)$ coincides with the support of
2106.07890_F00245	20	■ 1.000	$G, H: \mathcal{C} \rightarrow$	$G, H: \mathcal{C} \rightarrow$	internal hom between a pair of ordinary copresheaves $G, H: \mathcal{C} \rightarrow \text{Set}$ is the co-
2106.07890_F00246	20	■ 1.000	$c \mapsto \widehat{\mathcal{C}}(h^c \times G, H)$	$c \mapsto \widehat{\mathcal{C}}(h^c \times G, H)$	presheaf $[G, H]$ defined on objects by $c \mapsto \widehat{\mathcal{C}}(h^c \times G, H)$, which is the set of
2106.07890_F00247	20	■ 1.000	$h^c \times G$	$h^c \times G$	natural transformations from $h^c \times G$ to H .
2106.07890_F00248	20	■ 1.000	$[f, g]: \mathcal{L} \rightarrow [0, 1]$	$[f, g]: \mathcal{L} \rightarrow [0, 1]$	Definition 12. For any $f, g: \mathcal{L} \rightarrow [0, 1]$ let $[f, g]: \mathcal{L} \rightarrow [0, 1]$ be defined by

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2106.07890_F00249	20	■ 1.000	$h^{\{c\}} \times f$	$h^c \times f$	where $\boxed{h^c \times f}$ is the product defined above Lemma 4.
2106.07890_F00250	20	■ 1.000	$\mathcal{L}(c, d) \leq [[f, g](c), [f, g](d)]$	$\mathcal{L}(c, d) \leq [[f, g](c), [f, g](d)]$	<i>Proof.</i> We need to show that $\boxed{\mathcal{L}(c, d) \leq [[f, g](c), [f, g](d)]}$, which is equivalent to
2106.07890_F00251	20	■ 1.000	$\mathcal{L}(c, d)[f, g](c) \leq [f, g](d)$	$\mathcal{L}(c, d)[f, g](c) \leq [f, g](d)$	showing that $\boxed{\mathcal{L}(c, d)[f, g](c) \leq [f, g](d)}$. Start with the fact that g is a copresheaf
2106.07890_F00252	20	■ 1.000	$\mathcal{L}(c, d) \leq [g(c), g(d)]$	$\mathcal{L}(c, d) \leq [g(c), g(d)]$	to get $\mathcal{L}(c, d) \leq [g(c), g(d)]$ which is equivalent to $\mathcal{L}(c, d)g(c) \leq g(d)$ which by
2106.07890_F00253	20	■ 1.000	$\mathcal{L}(c, d) g(c) \leq g(d)$	$\mathcal{L}(c, d)g(c) \leq g(d)$	to get $\mathcal{L}(c, d) \leq [g(c), g(d)]$ which is equivalent to $\boxed{\mathcal{L}(c, d)g(c) \leq g(d)}$ which by
2106.07890_F00254	20	■ 1.000	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c, g) \leq \widehat{\mathcal{L}}(h^d, g)$	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c, g) \leq \widehat{\mathcal{L}}(h^d, g)$	the enriched Yoneda Lemma is equivalent to $\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c, g) \leq \widehat{\mathcal{L}}(h^d, g)$. This is,
2106.07890_F00255	20	■ 1.000	$\mathcal{L}(c, d) \left(\widehat{\mathcal{L}}(h^c, g) \times \widehat{\mathcal{L}}(f, g) \right) \leq \widehat{\mathcal{L}}(h^d, g) \times \widehat{\mathcal{L}}(f, g)$	$\mathcal{L}(c, d) \left(\widehat{\mathcal{L}}(h^c, g) \times \widehat{\mathcal{L}}(f, g) \right) \leq \widehat{\mathcal{L}}(h^d, g) \times \widehat{\mathcal{L}}(f, g)$	in turn, equivalent to $\boxed{\mathcal{L}(c, d) \left(\widehat{\mathcal{L}}(h^c, g) \times \widehat{\mathcal{L}}(f, g) \right) \leq \widehat{\mathcal{L}}(h^d, g) \times \widehat{\mathcal{L}}(f, g)}$. Using
2106.07890_F00256	20	■ 1.000	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c \times f, g) \leq \widehat{\mathcal{L}}(h^d \times f, g)$	$\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c \times f, g) \leq \widehat{\mathcal{L}}(h^d \times f, g)$	Lemma 4, we can rewrite this inequality as $\boxed{\mathcal{L}(c, d) \widehat{\mathcal{L}}(h^c \times f, g) \leq \widehat{\mathcal{L}}(h^d \times f, g)}$,
2106.07890_F00257	21	■ 1.000	$\mathcal{L}(c, d) \leq \left[\widehat{\mathcal{L}}(h^c \times f, g), \widehat{\mathcal{L}}(h^d \times f, g) \right] = [[f, g](c), [f, g](d)]$	$\mathcal{L}(c, d) \leq \left[\widehat{\mathcal{L}}(h^c \times f, g), \widehat{\mathcal{L}}(h^d \times f, g) \right] = [[f, g](c), [f, g](d)]$	which is equivalent to $\boxed{\mathcal{L}(c, d) \leq [\widehat{\mathcal{L}}(h^c \times f, g), \widehat{\mathcal{L}}(h^d \times f, g)] = [[f, g](c), [f, g](d)]}$
2106.07890_F00258	21	■ 1.000	$(f \times -) : \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}$	$(f \times -) : \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}$	We have an enriched functor $\boxed{f \times - : \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}}$ and Definition 12 presents $[f, -]$
2106.07890_F00259	21	■ 1.000	$[f, -]$	$[f, -]$	We have an enriched functor $(f \times -) : \widehat{\mathcal{L}} \rightarrow \widehat{\mathcal{L}}$ and Definition 12 presents $[f, -]$
2106.07890_F00260	21	■ 1.000	$- \times f$	$- \times f$	which is enriched right adjoint to $- \times f$. The enriched Yoneda Lemma (Theorem 1)
2106.07890_F00261	21	■ 1.000	$\widehat{\mathcal{L}}(h^c, [f, g]) = [f, g](c)$	$\widehat{\mathcal{L}}(h^c, [f, g]) = [f, g](c)$	says $\widehat{\mathcal{L}}(h^c, [f, g]) = [f, g](c)$ and by definition, we have $[f, g](c) = \widehat{\mathcal{L}}(h^c \times f, g)$.
2106.07890_F00262	21	■ 1.000	$[f, g](c) = \widehat{\mathcal{L}}(h^c \times f, g)$	$[f, g](c) = \widehat{\mathcal{L}}(h^c \times f, g)$	says $\widehat{\mathcal{L}}(h^c, [f, g]) = [f, g](c)$ and by definition, we have $\boxed{[f, g](c) = \widehat{\mathcal{L}}(h^c \times f, g)}$.
2106.07890_F00263	21	■ 1.000	$\widehat{\mathcal{L}}(h^c, [f, g]) = \widehat{\mathcal{L}}(h^c \times f, g)$	$\widehat{\mathcal{L}}(h^c, [f, g]) = \widehat{\mathcal{L}}(h^c \times f, g)$	The equality $\widehat{\mathcal{L}}(h^c, [f, g]) = \widehat{\mathcal{L}}(h^c \times f, g)$ for all representables h^c implies equality
2106.07890_F00264	21	■ 1.000	$h^{\{c\}}$	h^c	The equality $\widehat{\mathcal{L}}(h^c, [f, g]) = \widehat{\mathcal{L}}(h^c \times f, g)$ for all representables h^c implies equality
2106.07890_F00265	21	■ 1.000	h	h	for all copresheaves h :
2106.07890_F00266	21	■ 1.000	$[f, g]$	$[f, g]$	That is, $[f, g]$ serves as an internal-hom for $\widehat{\mathcal{L}}$ making it enriched <i>Cartesian closed</i> .
2106.07890_F00267	21	■ 1.000	$x, y \in \mathcal{L}$	$x, y \in \mathcal{L}$	Theorem 4. For any $x, y \in \mathcal{L}$, we have
2106.07890_F00268	21	■ 0.999	$\left[h^x, h^y \right](c) =$	$[h^x, h^y](c) =$	Note that if there is a text c that contains x but does not contain y , then $\boxed{[h^x, h^y](c) =}$
2106.07890_F00269	21	■ 1.000	$d = c$	$d = c$	0 for the infimum in Equation (21) is realized when $d = c$: the numerator $\pi(c y) =$
2106.07890_F00270	21	■ 1.000	$\pi(c \mid y) =$	$\pi(c \mid y) =$	0 for the infimum in Equation (21) is realized when $d = c$: the numerator $\pi(c y) =$
2106.07890_F00271	21	■ 1.000	$\pi(c \mid c) \times \pi(c \mid x) = \pi(c \mid x) \neq 0$	$\pi(c \mid c) \times \pi(c \mid x) = \pi(c \mid x) \neq 0$	0 and the denominator $\pi(c c) \times \pi(c x) = \pi(c x) \neq 0$. So, if $[h^x, h^y](c) \neq 0$,
2106.07890_F00272	21	■ 1.000	$\left[h^x, h^y \right](c) \neq 0$	$[h^x, h^y](c) \neq 0$	0 and the denominator $\pi(c c) \times \pi(c x) = \pi(c x) \neq 0$. So, if $[h^x, h^y](c) \neq 0$,
2106.07890_F00273	21	■ 1.000	$x \Rightarrow y$	$x \Rightarrow y$	Definition 13. For any expressions $x, y \in \mathcal{L}$, define the implication $x \Rightarrow y$ to be
2106.07890_F00274	21	■ 1.000	$\left[h^x, h^y \right] : \mathcal{L} \rightarrow [0, 1]$	$[h^x, h^y] : \mathcal{L} \rightarrow [0, 1]$	the copresheaf $\boxed{[h^x, h^y] : \mathcal{L} \rightarrow [0, 1]}$.
2106.07890_F00275	21	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	It is sometimes desirable to work in a category $\mathcal{M}$ that is a slight variant of the
2106.07890_F00276	21	■ 0.999	$[0, \infty]$	$[0, \infty]$	to each hom object. First notice the set of nonnegative extended reals $[0, \infty]$ to-
2106.07890_F00277	21	■ 1.000	$a + \infty := \infty$	$a + \infty := \infty$	$a + \infty := \infty$ and $\infty + a := \infty$ for all a . If one further specifies a morphism
2106.07890_F00278	21	■ 1.000	$\infty + a := \infty$	$\infty + a := \infty$	$a + \infty := \infty$ and $\infty + a := \infty$ for all a . If one further specifies a morphism
2106.07890_F00279	21	■ 0.998	$b \leq a$	$b \leq a$	from a to b whenever $b \leq a$, then $[0, \infty]$ is a commutative monoidal preorder.

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2106.07890_F00280	22	—	$d_{\mathcal{M}}$	$d_{\mathcal{M}}$	
2106.07890_F00281	22	■ 1.000	$[a, b] := \max\{b - a, 0\}$	$[a, b] := \max\{b - a, 0\}$	$[a, b] := \max\{b - a, 0\}$, and the limit of any diagram is <i>supremum</i> of the num-
2106.07890_F00282	22	■ 1.000	$a \times b = \max\{a, b\}$	$a \times b = \max\{a, b\}$	categorical product and coproduct are respectively given by $a \times b = \max\{a, b\}$
2106.07890_F00283	22	■ 0.997	$a \sqcup b = \min\{a, b\}$	$a \sqcup b = \min\{a, b\}$	and $a \sqcup b = \min\{a, b\}$ for all a and b in $[0, \infty]$. The function $-\ln: [0, 1] \rightarrow [0, \infty]$
2106.07890_F00284	22	■ 0.997	$-\ln: [0, 1] \rightarrow [0, \infty]$	$-\ln: [0, 1] \rightarrow [0, \infty]$	and $a \sqcup b = \min\{a, b\}$ for all a and b in $[0, \infty]$. The function $-\ln: [0, 1] \rightarrow [0, \infty]$
2106.07890_F00285	22	■ 1.000	$[0, \infty] \rightarrow [0, 1]$	$[0, \infty] \rightarrow [0, 1]$	$[0, \infty] \rightarrow [0, 1]$ is the map $a \mapsto \exp(-a)$, and both maps are continuous and
2106.07890_F00286	22	■ 1.000	$a \mapsto \exp(-a)$	$a \mapsto \exp(-a)$	$[0, \infty] \rightarrow [0, 1]$ is the map $a \mapsto \exp(-a)$, and both maps are continuous and
2106.07890_F00287	22	■ 1.000	$\mathcal{M}(x, y) := -\ln \mathcal{L}(x, y)$	$\mathcal{M}(x, y) := -\ln \mathcal{L}(x, y)$	$\mathcal{M}(x, y) := -\ln \mathcal{L}(x, y)$. It is then straightforward to check that $0 \geq \mathcal{M}(x, x)$
2106.07890_F00288	22	■ 1.000	$0 \geq \mathcal{M}(x, x)$	$0 \geq \mathcal{M}(x, x)$	$\mathcal{M}(x, y) := -\ln \mathcal{L}(x, y)$. It is then straightforward to check that $0 \geq \mathcal{M}(x, x)$
2106.07890_F00289	22	■ 1.000	$\mathcal{M}(x, y) + \mathcal{M}(y, z) \geq \mathcal{M}(x, z)$	$\mathcal{M}(x, y) + \mathcal{M}(y, z) \geq \mathcal{M}(x, z)$	for all expressions x and moreover that $\mathcal{M}(x, y) + \mathcal{M}(y, z) \geq \mathcal{M}(x, z)$ for all
2106.07890_F00290	22	■ 0.998	$d_{\mathcal{M}}(x, y) := \mathcal{M}(x, y)$	$d_{\mathcal{M}}(x, y) := \mathcal{M}(x, y)$	in $\mathcal{M}$ by $d_{\mathcal{M}}(x, y) := \mathcal{M}(x, y)$, thinking of $d_{\mathcal{M}}$ as defining a kind of distance
2106.07890_F00291	22	■ 1.000	$\widehat{\mathcal{M}} := [0, \infty]^{\mathcal{M}}$	$\widehat{\mathcal{M}} := [0, \infty]^{\mathcal{M}}$	plies in $\mathcal{M}$ as well. There is a generalized metric space $\widehat{\mathcal{M}} := [0, \infty]^{\mathcal{M}}$ whose
2106.07890_F00292	23	■ 1.000	$f: \mathcal{M} \rightarrow [0, \infty]$	$f: \mathcal{M} \rightarrow [0, \infty]$	objects are $[0, \infty]$ -coplesheaves on $\mathcal{M}$, which are functions $f: \mathcal{M} \rightarrow [0, \infty]$ satis-
2106.07890_F00293	23	■ 1.000	$d_{\mathcal{M}}(x, y) \geq [fx, fy] = \max\{fy - fx, 0\}$	$d_{\mathcal{M}}(x, y) \geq [fx, fy] = \max\{fy - fx, 0\}$	fying $d_{\mathcal{M}}(x, y) \geq [fx, fy] = \max\{fy - fx, 0\}$ for all expressions x and y in the
2106.07890_F00294	23	■ 1.000	$[a, b] = \max\{b - a, 0\}$	$[a, b] = \max\{b - a, 0\}$	$[a, b] = \max\{b - a, 0\}$ by $d_{[0, \infty]}(a, b)$. So a copresheaf is like a metric contraction;
2106.07890_F00295	23	■ 1.000	$d_{[0, \infty]}(a, b)$	$d_{[0, \infty]}(a, b)$	$[a, b] = \max\{b - a, 0\}$ by $d_{[0, \infty]}(a, b)$. So a copresheaf is like a metric contraction;
2106.07890_F00296	23	■ 1.000	$d_{[0, \infty]}(fx, fy) \leq d_{\mathcal{M}}(x, y)$	$d_{[0, \infty]}(fx, fy) \leq d_{\mathcal{M}}(x, y)$	that is, a function $f: \mathcal{M} \rightarrow [0, \infty]$ satisfying $d_{[0, \infty]}(fx, fy) \leq d_{\mathcal{M}}(x, y)$.
2106.07890_F00297	23	■ 1.000	$\widehat{\mathcal{M}}$	$\widehat{\mathcal{M}}$	space $\widehat{\mathcal{M}}$ is also both complete and cocomplete with respect to weighted (co)limits,
2106.07890_F00298	23	■ 0.905	$\mathcal{M}^{\text{op}} \rightarrow \widehat{\mathcal{M}}$	$\mathcal{M}^{\text{op}} \rightarrow \widehat{\mathcal{M}}$	and moreover the Yoneda embedding $\mathcal{M}^{\text{op}} \rightarrow \widehat{\mathcal{M}}$ that maps $x \mapsto d_{\mathcal{M}}(x, -)$ de-
2106.07890_F00299	23	■ 0.905	$x \mapsto d_{\mathcal{M}}(x, -)$	$x \mapsto d_{\mathcal{M}}(x, -)$	and moreover the Yoneda embedding $\mathcal{M}^{\text{op}} \rightarrow \widehat{\mathcal{M}}$ that maps $x \mapsto d_{\mathcal{M}}(x, -)$ de-
2106.07890_F00300	23	■ 0.998	$f, g \in \widehat{\mathcal{M}}$	$f, g \in \widehat{\mathcal{M}}$	Theorem 5. For any copresheaves $f, g \in \widehat{\mathcal{M}}$ and any weights w_1, w_2 , the weighted
2106.07890_F00301	23	■ 0.998	w_1, w_2	w_1, w_2	Theorem 5. For any copresheaves $f, g \in \widehat{\mathcal{M}}$ and any weights w_1, w_2 , the weighted
2106.07890_F00302	23	■ 1.000	$f, g: \mathcal{M} \rightarrow [0, \infty]$	$f, g: \mathcal{M} \rightarrow [0, \infty]$	<i>Proof.</i> We translate the results from Sections 4.2 and 4.3. Let $f, g: \mathcal{M} \rightarrow [0, \infty]$
2106.07890_F00303	23	■ 1.000	$w_1, w_2 \in [0, \infty]$	$w_1, w_2 \in [0, \infty]$	and $w_1, w_2 \in [0, \infty]$. Define $f', g': \mathcal{L} \rightarrow [0, 1]$ and weights $w'_1, w'_2 \in [0, 1]$ by
2106.07890_F00304	23	■ 1.000	$f', g': \mathcal{L} \rightarrow [0, 1]$	$f', g': \mathcal{L} \rightarrow [0, 1]$	and $w_1, w_2 \in [0, \infty]$. Define $f', g': \mathcal{L} \rightarrow [0, 1]$ and weights $w'_1, w'_2 \in [0, 1]$ by
2106.07890_F00305	23	■ 1.000	$w'_1, w'_2 \in [0, 1]$	$w'_1, w'_2 \in [0, 1]$	and $w_1, w_2 \in [0, \infty]$. Define $f', g': \mathcal{L} \rightarrow [0, 1]$ and weights $w'_1, w'_2 \in [0, 1]$ by
2106.07890_F00306	23	■ 1.000	$f'c = \exp(-fc), g'c = \exp(-gc)$	$f'c = \exp(-fc), g'c = \exp(-gc)$	setting $f'c = \exp(-fc), g'c = \exp(-gc)$, and $w'_i = \exp(-w_i)$ for $i = 1, 2$. Then
2106.07890_F00307	23	■ 1.000	$w'_i = \exp(-w_i)$	$w'_i = \exp(-w_i)$	setting $f'c = \exp(-fc), g'c = \exp(-gc)$, and $w'_i = \exp(-w_i)$ for $i = 1, 2$. Then
2106.07890_F00308	24	■ 1.000	$d_{\mathcal{M}}(x, -)$	$d_{\mathcal{M}}(x, -)$	copresheaves $d_{\mathcal{M}}(x, -)$ suggests that the categorical completion $\widehat{\mathcal{M}}$ resembles a
2106.07890_F00309	24	■ 0.968	$((-\infty, \infty], \oplus, \odot)$	$((-\infty, \infty], \oplus, \odot)$	Definition 14. The data $((-\infty, \infty], \oplus, \odot)$ with operations $\oplus$ and $\odot$ defined by
2106.07890_F00310	24	■ 1.000	$([0, \infty], \oplus, \odot)$	$([0, \infty], \oplus, \odot)$	semi-ring $([0, \infty], \oplus, \odot)$ is called the <i>semi-tropical semi-ring</i> . A module over a
2106.07890_F00311	24	■ 0.999	$w_1 = w_2 = 0$	$w_1 = w_2 = 0$	Theorem 6. The coproduct (with trivial weights $w_1 = w_2 = 0$) in $\widehat{\mathcal{M}}$ makes it into
2106.07890_F00312	24	■ 0.999	$[0, \infty] \times \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}$	$[0, \infty] \times \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}$	a commutative monoid. The map $[0, \infty] \times \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}$ defined by $(s, f) \mapsto s \odot f$
2106.07890_F00313	24	■ 0.999	$(s, f) \mapsto s \odot f$	$(s, f) \mapsto s \odot f$	a commutative monoid. The map $[0, \infty] \times \widehat{\mathcal{M}} \rightarrow \widehat{\mathcal{M}}$ defined by $(s, f) \mapsto s \odot f$
2106.07890_F00314	24	■ 0.775	$(s \odot f)(x) = f(x) + s$	$(s \odot f)(x) = f(x) + s$	where $(s \odot f)(x) = f(x) + s$ makes $\mathcal{M}$ into a module over the semi-tropical
2106.07890_F00315	24	■ 1.000	$\left(s_1 \oplus s_2\right) f(x) = \min\left\{f(x) + s_1, f(x) + s_2\right\}$	$(s_1 \oplus s_2) f(x) = \min\{s_1, s_2\} f(x) = \min\{f(x) + s_1, f(x) + s_2\}$	<i>Proof.</i> We check: $(s_1 \oplus s_2) f(x) = \min\{s_1, s_2\} f(x) = \min\{f(x) + s_1, f(x) + s_2\}$

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2106.07890_F00316	24	■ 1.000	$\left(s_{\{1\}} \odot s_{\{2\}}\right) f(x)=f x+s_{\{1\}}+s_{\{2\}}$	$\left(s_1 \odot s_2\right) f(x)=f x+s_1+s_2$	and $\boxed{\left(s_1 \odot s_2\right) f(x)=f x+s_1+s_2}$ which is the same as $\left(s_1\right) \odot\left(s_2 \odot f\right)(x)=$
2106.07890_F00317	24	■ 1.000	$\left(s_{\{1\}}\right) \odot\left(s_{\{2\}} \odot f\right)(x)=$	$\left(s_1\right) \odot\left(s_2 \odot f\right)(x)=$	and $\left(s_1 \odot s_2\right) f(x)=f x+s_1+s_2$ which is the same as $\boxed{\left(s_1\right) \odot\left(s_2 \odot f\right)(x)=}$
2106.07890_F00318	24	■ 1.000	$\left(s_{\{2\}} \odot f\right)(x)+s_{\{1\}}=f x+s_{\{2\}}+s_{\{1\}}$	$\left(s_2 \odot f\right)(x)+s_1=f x+s_2+s_1$	$\boxed{\left(s_2 \odot f\right)(x)+s_1=f x+s_2+s_1}$.
2106.07890_F00319	26	■ 0.934	$\sqcup$	$\sqcup$	guage, but rather captures the concept “he” $\sqcup$ “she.” Density operators also have